Chapter 13, Problem 7RQ

Problem

A three-winding transformer is connected as portrayed in Fig. 13.71(a). The value of the output voltage *Vo* is:

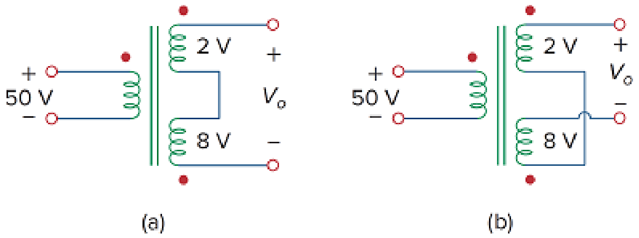
(a) 10

(b) 6

(c) −6

(d) −10

Figure 13.71



Step-by-step solution

Step 1 of 2

Refer to Figure 13.71 (a) in the textbook.

The polarities of the two windings of the secondary side are same when it is series aiding.

From the Figure, the current enters through one dot and leaves through other dot.

The voltages are,

$$V_1 = 2\text{ V}$$
$$V_2 = -8\text{ V}$$

Step 2 of 2

Calculate the output voltage, V_o.

$$V_o = V_1 + V_2$$

Substitute 2 V for V_1, and -8 V for V_2

$$V_o = 2 + (-8) = -6\text{ V}$$

Thus, the correct answer is $\boxed{c}$.